Time Separation, Coordination, and Performance in Technical Teams

J. Alberto Espinosa, *Member, IEEE*, Jonathon N. Cummings, and Cynthia Pickering

***Abstract*—Technical teams are often distributed across geographic locations and across time zones. While spatial and time separation are often correlated, most prior studies have only focused on one or the other. As a consequence, their respective effects may be confounded when teams have both spatial and time separation. We argue that bridging spatial and time separation pose very different coordination challenges, thus their respective impacts need to be examined together to fully understand how geographic configuration influences team performance. We report on a field study of 123 technical teams conducted at a large semiconductor manufacturing company where we investigated how spatial and time separation influenced team performance. Our results show that time separation, in the form of maximum time zone difference spanned by members, has a stronger negative impact on team performance than spatial separation. We also show that this impact is indirect, i.e., large time zone spans create coordination problems, which in turn impact team performance. Put differently, when coordination problems are reduced, the negative association between maximum time zone span and performance disappears. We describe our findings and discuss implications for global team managers and collaboration tool designers.**

***Index Terms*—Business, collaboration, engineering management, global communication, globalization, international collaboration, innovation management, knowledge management, management, organizational aspects, research and development management, technology management, teamwork, virtual groups.**

I. Introduction

GEOGRAPHICALLY dispersed technical teams have been at the forefront of globalization efforts. As a recent report by IBM stated, *"companies continue to expand globally, distributing their teams around the world through a variety of means, including offshoring, acquiring, partnering, and outsourcing. As globalization becomes more prevalent, many companies are evolving their approach and practices, perhaps demonstrating a maturation of the distributed model"* [1]. This trend is likely to accelerate at an even greater pace in the coming years, at least in part because of the demand for technical teams spread across geographic locations and time zones. For example, a recent Gartner report [2] estimates that the growth of globally delivered IT services (10%-30%) will surpass the growth of IT services as a whole (7%-8%), going from approximately $70 billion or 9% of the total IT services market in 2009, to a projected $120-$150 billion (or 13%-16.5% of the market) in 2013.

This ever-increasing demand for the development and delivery of global products and services puts a lot of pressure on technical teams to achieve high levels of performance. Performance in technical teams is often conceptualized along two key dimensions: process performance and product performance [3]–[5]. Process performance refers to how well a project was undertaken by the team. It is often evaluated by measures like on-time and on-budget completion of the project, user participation and team member satisfaction among other things [4], [6], [7]. Product performance refers to the effectiveness of the product or service implemented by the team, and it is generally evaluated by measures like system quality, hardware and software functionality and their net benefits, among other things [8]. Consistent with this research, we conceptualize technical team performance in this study as the process and product success, evidenced by its on-time completion within-budget and meeting system requirements [9].

Because technical tasks are generally complex, involving highly interdependent collaborative activities carried out by multiple individuals, good coordination is a necessary condition for high performance. Coordination is defined as the "management of dependencies among task activities" [10]. Team members not only need to be productive individually, but also need to work with each other to manage their interdependencies effectively. This conceptualization of coordination is very useful, because, it allows us to define constructs more precisely and distinguish coordination from other related concepts like collaboration and integration[1]. It also helps distinguish between the act of coordinating—i.e., the actions undertaken to manage dependencies among task activities—and coordination outcomes—i.e., the extent to which dependencies were well managed—as evidenced by fewer coordination problems.

Coordination is very important for performance in technical tasks, but is a very challenging undertaking when the work is

Manuscript received May 31, 2010; revised October 1, 2010, and December 22, 2010; accepted February 18, 2011. This project was supported by the American University's Center for Information Technology and the Global Economy (CITGE), National Science Foundation (NSF) Award No. IIS-0603667, and a research grant from Intel Corporation. Review of this manuscript was arranged by Department Editor Jeffrey K. Pinto.

J. A. Espinosa is with American University, 4400 Massachusetts Ave., N.W., Washington, DC 20016 USA (e-mail: alberto@american.edu).

J. N. Cummings is with Duke University, Durham, NC 27705 USA (e-mail: jonathon.cummings@duke.edu).

C. Pickering is with Intel Corporation, Phoenix, AZ 285016 USA (e-mail: cynthia.k.pickering@intel.com).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TEM.2011.2126579

[1]Collaboration is simply working together. Effective collaborations may or may not need coordination depending on the nature of interdependencies among task activities. Collaboration in tasks with relatively independent activities may not benefit as much from coordination, and it may actually be costly because unnecessary coordination diverts attention from the productive task. Integration broadly defined means combining parts into a unitary whole. Naturally, if the parts are interdependent, getting them to interoperate effectively will require some coordination.

carried out across geographic boundaries. Research in geographically distributed teams has categorized the communication technologies that members use and how they interact based on whether they collaborate in the same (i.e., collocated) or different place (i.e., spatially separated), and in the same (i.e., synchronous) or different time (i.e., asynchronous or time separated) [11], [12]. Therefore, spatial separation due to geographic distance and time separation due to time zone differences are two key geographic characteristics of the work context that can have a substantial impact on how team members work together. The additional coordination overhead imposed by spatial and time separation may divert too much attention from the individual production tasks, therefore, offsetting some of the cost advantages sought with global technical work. However, research has not adequately distinguished the effects of spatial separation from time separation.

While there has been a lot of empirical research on coordination and performance of geographically dispersed teams [13]–[18], most studies either have not specifically addressed the impact of time separation or have treated it as a discussion item. As Olson and Olson [18] concluded some time ago from their review of research in this area, spatial separation per se may not matter that much in teamwork, but other factors like time separation may be what makes a difference. Studies often treat time separation as a consequence of spatial separation, and one may be quick to conclude that time separation cannot occur without some form of spatial separation. It is precisely because spatial and time separation are often associated with each other that they should be studied together [19], [20]. We elaborate on this point next.

Except for teams dispersed in a North–South geographic configuration, spatial and time separation tend to correlate statistically. Despite this correlation, bridging these two forms of separation poses very different challenges. For example, Allen studied how distance affected collaboration in engineering organizations and found that communication frequency dropped substantially when collaborators were only a few meters apart [21]. Similarly, Kraut and colleagues found that there was less collaboration between scientists when they were separated by distance [22]. However, in today's digital age, some problems associated with spatial separation can be addressed with real-time technology, e.g., telephone, video or web conference, texting, or instant messaging. For example, Sobel–Lojeski conducted interviews with digital age leaders and concluded that "techno-dexterity" helps collaborators communicate effectively, reducing the "perceived" distance among them [23].

Furthermore, once members are separated spatially, the actual magnitude of separation is less likely to affect how the team interacts. As Allen found in the study we discussed above, once collaborators were separated a few meters, their communication frequency did not drop much more with further spatial separation [21]. In essence, some key aspects of spatial separation (without time separation) include that: 1) spatial separation eliminates some of the benefits of co-presence, but some of these benefits can be effectively restored through communication technologies; 2) work hours fully overlap, so team members can chose between synchronous (i.e., same-time) and asynchronous (i.e., different-time) communication modes based on what they feel is most effective for the situation at hand; and 3) the magnitude of the distance does not affect the team's interaction (e.g., working across 200 miles compared with 2000 miles).

In contrast, the amount of work–time overlap, the magnitude of time differences, and the dispersion of members across time zones do matter with time separation. Small time zone differences may not dramatically alter how members interact, and tactics to reduce time separation like shifting work hours, flextime, and telecommunication may be sufficient. However, when team members are located in multiple time zones, especially when the time zone difference spanned is large, simple things like coordinating meetings, scheduling phone calls, and locating colleagues become daunting tasks, as this comment made by a global team member we interviewed illustrates: *"there are about four technical people that we meet with every morning at 5:00 a.m. Pacific time to review what progress they have made and any issues that we have seen from the previous day. Then we meet with our end users at 8:30 [a. m.] and tell them what to test that day and any feedback from the technical team, and then we meet with the end users again at 3:30 in the afternoon to get the test results and any feedback that they wanted us to give to the technical team the next morning."*

This team member's comment suggests that in contrast to spatial separation, time separation can dictate the level of synchronicity of the workflow and alter the timing of team member interactions [24]. Furthermore, as the time zone difference spanned by the team increases, the overlapping window of opportunity for synchronous interaction decreases, further restricting the timing options. As we have argued, prior studies of global teams have not clearly differentiated between spatial and time separation [25]. Our study fills this gap.

The key distinguishing feature of time separation is that it breaks up the work day into two distinct components, i.e., overlapping time and nonoverlapping time. As the overlapping time is reduced, the window of opportunity for synchronous interaction diminishes. This forces the team into more asynchronous collaboration, thus exacerbating the problems associated with lack of copresence. For example, in a recent study conducted at Infosys in India, Carmel [26] suggested that when coordination breaks down, work often needs to stop until the next day when colleagues on the other site come back to work. He observed that Infosys has become effective at working across time zones, not by eliminating the associated problems of time separation, but by learning to manage these problems and make them predictable, resorting to workarounds like work–time flexibility, a 24-hour work environment, and well defined processes and escalation protocols, among other things. Outsourcing developers in India have the benefit that most clients are in a given time zone, but when team members operate in various time zones, the overlapping time between each pair of sites varies substantially, and the amount of simultaneous overlapping time for all sites involved narrows dramatically. Finally, having to constantly switch from synchronous to asynchronous interaction and keeping track of when this switch needs to be done for each location can be cognitively costly.

As our discussion above suggests, while spatial and time separation often correlate statistically—particularly when dispersed

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

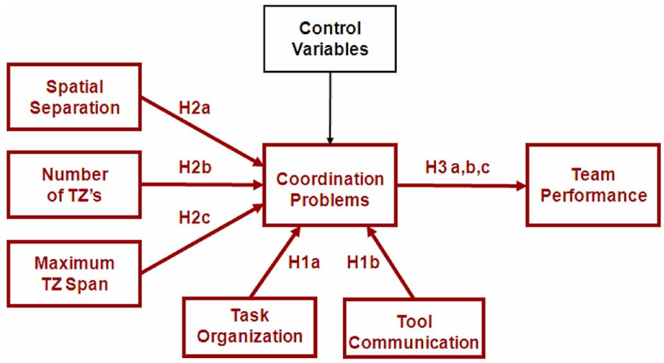

Fig. 1. Path model.

East to West—they are conceptually distinct. Thus, it is very important to address the specific challenges of each so that solutions to bridge both types of separation can be more effectively developed. Unless all team members are in the same time zone, not controlling for time separation variables may confound the effects of spatial separation (and vice versa) because of omitted variable bias (i.e., the included variable picks up some of the effects of the omitted variable). In contrast, when both variables are included, the coefficients are unbiased. We argue that if the study context includes teams with both spatial and time separation, it is important to include both types of variables in the study in order to avoid omitted variable bias.

In sum, the specific effects of spatial and time separation on team coordination and performance have not been adequately distinguished in the literature, which provides the foundation for our study, i.e., teasing out how each type of separation affects team coordination and performance, while controlling for the other. The primary goal of this study is to better understand the effects of time separation on team coordination and team performance, and to statistically differentiate them from the effects of spatial separation. The primary factors we analyze in our study are spatial separation among team members and two aspects of time separation—number of time zones and maximum time zone differences spanned by the team (e.g., a team with members in New York, the U. K., and India, operates in three time zones and has a span of 10.5 hours)—to better understand how different configuration parameters affect team coordination and team performance. We also account for other factors that can affect team coordination and team performance, like team size, project length, project priority, project resources, and communication tool use. In the next section, we provide our theoretical foundations and hypotheses. We then describe our study methods, followed by our data analysis and results. Finally, we offer a discussion of results, limitations, and concluding remarks.

II. Theoretical Development and Hypotheses

Consistent with prior research on teams, we specify our research model as a three-stage path model [27]: 1) antecedents of coordination problems, 2) coordination problems as a mediator, and 3) team performance outcomes. Our research model is illustrated in Fig. 1.

A. Antecedents of Coordination Problems

There has been a substantial amount of research over the years on the antecedents of coordination [28], [29], so we only offer two associated hypotheses for completeness of our research model. Classical organizational theory suggests that team members coordinate their work "mechanistically" via task organization and "organically" through communication [30], [31]. Mechanistic coordination, i.e., task organization, is most effective for routine aspects of the task that can be programmed and it involves things like role assignment, division of labor, ground rules and routines. Prior research has found that certain tools specifically designed to provide some form of task organization (e.g., software configuration management systems) help teams work more effectively [32]. Thus, we posit:

> Hypothesis 1a: Task organization is negatively associated with coordination problems in technical teams.

Organic coordination or coordination by mutual adjustment, i.e., team communication or feedback, is most effective for less routine aspects. Prior research has provided ample evidence that communication is important for coordination with software teams [16], [33] and other work teams [29]. Thus, we posit:

> Hypothesis 1b: Team communication is negatively associated with coordination problems in technical teams.

B. Geographic Configuration

Geographically dispersed teams are separated by various team boundaries (e.g., spatial, temporal, functional, cultural, organizational) that make it more difficult for members to work together [34]–[36]. In this study, we focus specifically on two team boundaries associated with geographic dispersion, i.e., spatial and time separation, which are important in global work [11], [12]. Because team members can be distributed in a number of different ways across space and time, we refer to this concept as geographic configuration. How a team is configured geographically will affect its interaction dynamics [24]. For example, some teams may be entirely collocated, whereas some others may be spread out across multiple locations within the same or close time zones, yet others may be distributed among many time zones. Thus, the configuration of geographically dispersed teams varies depending on whether members are in the same location and time zone, multiple locations and time zones, or somewhere in between.

The extent to which a team is spatially separated will affect how a team coordinates its tasks, mainly because team member interaction needs to happen primarily through electronic media. At the same time, with limited or no time separation, team members can still have a substantial amount of synchronous interaction. In contrast, the number of time zones, in which the team operates, can make a big difference on how members coordinate their work, because it is easier to keep track of members in one or two time zones (e.g., New York and Bangalore) than those scattered across multiple time zones. Similarly, as the time zone spanned by the team increases, the window of opportunity for synchronous interaction narrows for the team, with the most

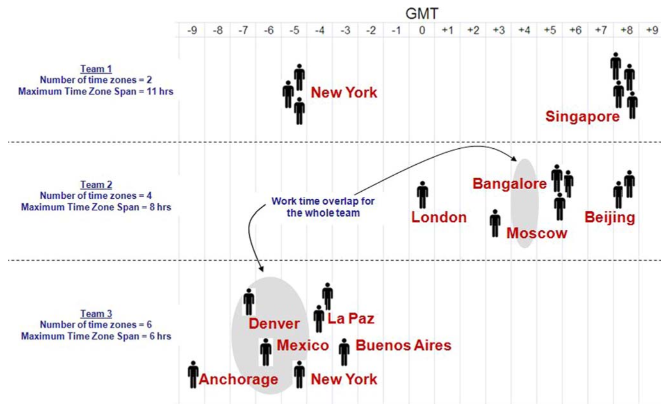

Fig. 2. Geographic configuration illustrations.

severe conditions happening when members have no work–time overlap, thus affecting how teams coordinate in these contexts.

In sum, we posit that it is not just spatial or time separation that makes a difference, but the actual way in which team members are configured geographically. Therefore, we study three geographic configuration variables of interest: spatial separation, number of time zones in which team members work, and the maximum time zone difference spanned by the team. We provide representative examples of various geographic configurations in Fig. 2. Note, how Team 1 has members in only two time zones, but has a maximum time zone span of 12 hours and no work–time overlap at all. In contrast, members in Team 3 are located in six different time zones, but their maximum time zone span is only six hours, leaving them three hours (in a nine hour work day) in which all members' work hours overlap. Team 2 is somewhere in between with four time zones represented, a maximum time zone span of eight hours, and a work–time overlap of one hour (for the entire team). We now discuss these variables in more detail and formulate hypotheses regarding their effect on team coordination and performance.

1) Spatial Separation: Researchers of global, virtual, and geographically dispersed teams have studied the effect of spatial separation, but most studies have treated it as a dichotomous variable—collocated or distributed. Prior work has shown that spatial separation can negatively impact coordination because it reduces the likelihood of face-to-face contact and spontaneous communication [21], [24], making it more difficult to resolve work issues [18], [37]. Spatial separation also reduces common ground [13] and hinders coordination [17]. However, most studies have not controlled for time separation and, therefore, we don't know if the reported effects can be attributed to spatial separation alone. Prior research has studied the effect of spatial separation on coordination delay (controlling for time zones) but has focused more specifically on delays within dyads [38], not on coordination at the team level. Sosa *et al.* [24] also studied the effect of spatial separation (controlling for time zones), but their focus was on communication frequency. Nevertheless, these two studies collectively suggest that even when controlling for time zones, spatial separation can still have an impact on coordination problems at the team level. Thus we posit:

> Hypothesis 2a: Controlling for the number of time zones and maximum time zone span in the team, spatial separation in technical teams is positively associated with coordination problems.

2) Time Separation: Time separation can introduce severe disruptions in the workflow among members, which can impact team coordination [39]. The timing of interaction is more critical for team members separated by time zones than those only separated by space alone. This is because members have to make choices between communicating asynchronously and waiting for overlapping work hours to communicate synchronously. In contrast to spatial separation, team interaction and timing can be affected by the magnitude of time separation. Moreover, the study of time separation brings about additional challenges that are very distinct from spatial separation. For example, whereas spatial separation can be measured through distance, travel time, and numbers of sites, time separation can be measured more precisely (e.g., number of time zones spanned, time zone differences between members).

In addition, the number of sites represented on a project within the same time zone does not severely affect the team interaction synchronicity, whereas the number of time zones can create severe workflow coordination challenges, so the timing of the interaction is critical [40]. As more time zones are represented in a team's geographic configuration, it becomes more difficult to plan activities and meetings because the collaboration context becomes more complex. As articulated in complexity theory, a task becomes more complex when the individuals involved in a task need to process more information cues to get the job done [41]. Studies have shown that when the complexity of the work context increases, team performance goes down [42].

We posit that such reductions in performance are due to the increased coordination problems resulting from the more complex context. For example, common tactics often used to overcome time separation, like schedule shifting and synchronous meetings during overlapping hours, become more problematic as more locations in different time zones are added. Shifting schedules may help two sites increase their work–time overlap, but may aggravate scheduling with a third site by narrowing the work–time overlap with that site. Similarly, planning meetings may be pretty straightforward when work hours overlap for the most part, but could be quite challenging when members are spread out across multiple time zones. Coordination challenges increase rapidly as more time zones are represented in a team, because the window for synchronous interaction is severely reduced and planning the timing of interaction becomes more difficult. Therefore, we posit that:

> Hypothesis 2b: Controlling for spatial separation and maximum time zone span in the team, the number of time zones represented in technical team is positively associated with coordination problems.

While spatial separation and the number of time zones in which the team operates may increase coordination challenges, the associated problems are less severe if the maximum time zone spanned by the team is small. This is because there is a wider window of opportunity for time shifting and synchronous interaction. Certain types of issues may be best resolved asynchronously, whereas others may be best dealt with synchronously. Therefore, when the window for synchronous interaction narrows, the choice of communication methods are substantially reduced to mostly asynchronous options, which may be effective for some aspects of the task that are not as sensitive to time, but inadequate for other aspects that may require more dynamic and frequent interaction. As the time zone span increases, this synchronous interaction window narrows considerably with the most severe cases occurring, when there are multiple sites without work–time overlap.

Similarly, the timing of workflows is more difficult when time zone spans are larger, introducing possible delays in work production as a consequence of having to wait until members in other sites come back to work from their prior day. A recent study found that the time zone difference between any two members is positively associated with their coordination delay and that such impact is more severe when there is no work–time overlap between them [38]. Furthermore, while communication tools helped alleviate the effect of spatial separation on coordination delay, none of the tools they examined had any such effect when the members had no work–time overlap. Such coordination delays will be further exacerbated when multiple pairs of members in the team have large time differences, thus causing substantial coordination problems. Therefore we posit that:

> Hypothesis 2c: Controlling for spatial separation and number of time zones represented in a team, the maximum time zone spanned by members of technical teams is positively associated with coordination problems.

C. Effect of Geographic Configuration on Team Performance, Mediated by Coordination Problems

We argue that geographic configuration influences team performance, but that this effect is mediated by coordination problems. Prior research suggests that when technical work is carried out across multiple locations, the results are missed deadlines and production delays [43]. Other studies have also noted the detrimental effects of geographic dispersion on team performance [13], [15], [16], [18]. We argue in this study that spatial and time separation present substantial coordination challenges to the team, and that it is the associated coordination problems that lead to diminished performance. Put differently, we argue that if the work can be effectively coordinated across spatial and time separation, then performance will be unaffected or the negative effects will be minimal.

With respect to spatial separation, prior research with software teams by Herbsleb and colleagues [44] found that teams working across geographical locations took longer to complete the task. But subsequent studies with similar data confirmed that spatial separation per se did not have a direct effect on task delay, but that teams working across distance required more team members for similar tasks, compared with collocated teams. Thus, when team size was incorporated into the path model, spatial separation was found to have an indirect effect on task delay mediated by team size. Herbsleb *et al.* speculated that spatially separated teams required larger teams to accommodated redundant roles on more than one site (e.g., more than one project manager, cross-site liaison personnel, etc.) to handle the additional coordination imposed by geographic distance.

Research studies, that have investigated the effect of time separation, have also been somewhat inconclusive. Some studies have shown that time separation can introduce delays in the task [38], but other studies have suggested that time differences can be used advantageously because some team members can advance the work while other team members go home at the end of the day [45]. The "follow-the-sun" (FTS) approach, i.e., work is handed from one site to another several time zones apart around the clock, suggests that development speed can be improved when the handoffs to the next site and the management of dependencies between sites can be effectively coordinated. However, the number of time zones in which team members are located is one substantial source of coordinative complexity, because members need to process more information to get the work done [41], thus creating more coordination problems, which in turn affects team performance.

Finally, as time separation increases, the handoffs from site to site become more problematic because team members have smaller windows for synchronous interaction. Furthermore, when the time separation is substantial, such that there is no work–time overlap, the result is increased coordination delay. Such coordination delay causes general coordination problems, which in turn affect performance negatively. Once again, if the task consists of relatively independent activities that require little coordination, the impact on team performance will be minimal [29]. Similarly, if activities are somewhat dependent but these dependencies can be effectively managed across large time separation spans, such that coordination problems are eliminated or substantially reduced, then performance will not be affected significantly [30], [31].

Together, the above arguments regarding spatial separation, time separation in the form of number of time zones, and time separation in the form of maximum time zone span suggest that:

> Hypothesis 3a: Spatial separation is positively associated with coordination problems, which in turn is negatively associated with technical team performance (i.e., an indirect effect and statistical mediation).

> Hypothesis 3b: Time separation in the form of number of time zones in a team is positively associated with coordination problems, which in turn is negatively associated with technical team performance (i.e., an indirect effect and statistical mediation).

> Hypothesis 3c: Time separation in the form of maximum time zone span is positively associated with coordination problems, which in turn is negatively associated with technical team performance (i.e., an indirect effect and statistical mediation).

III. Study Methods

We administered a web survey to team members working on technical projects at a semiconductor manufacturing company in 54 global locations across 23 countries in Europe, Asia, and South, Central, and North America. We targeted software and hardware engineers, but also included project managers,

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE I
DESCRIPTIVE STATISTICS AND CORRELATION MATRIX

	Variable	Mean	Std Dev	ICC	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Team Performance	3.74	0.60	0.19															
2	Coordination Problems	2.33	0.68	0.18	-0.64														
3	Team Size Core	10.21	6.31	N/A	-0.16	0.12													
4	Project Length in Months	28.44	25.46	N/A	0.02	-0.08	0.13												
5	Average Years with the Company	5.45	1.41	N/A	-0.05	-0.05	-0.20	0.15											
6	Project Resources	3.59	0.49	0.13	0.42	-0.39	0.03	0.01	-0.12										
7	Project Priority	3.88	0.65	0.17	0.05	0.22	0.01	-0.06	0.04	0.25									
8	Coordination via Task Organization	4.05	0.45	0.10	0.39	-0.59	-0.10	-0.02	0.05	0.49	0.17								
9	Coordination via Communication Tools	2.75	0.65	0.57	0.03	0.01	-0.51	-0.18	-0.14	-0.09	0.18	0.05							
10	Spatial Distance	4.83	1.29	N/A	0.00	-0.08	-0.08	0.03	0.33	-0.08	-0.20	0.02	-0.06						
11	Number of Time Zones	3.27	1.45	N/A	-0.08	0.08	0.48	0.23	0.07	-0.01	-0.04	-0.11	-0.40	0.40					
12	Max Time Zone Span	7.12	3.40	N/A	-0.20	0.20	0.19	0.08	-0.17	-0.02	0.00	-0.09	-0.19	0.20	0.44				
13	TZ Span >=0 and <3 hours	0.17	0.38	N/A	0.12	-0.14	-0.19	-0.05	0.01	-0.32	0.10	0.06	0.28	-0.30	-0.50	-0.86			
14	TZ Span >=3 and <6 hours	0.06	0.23	N/A	0.14	-0.14	-0.01	-0.04	0.17	0.12	-0.18	0.10	-0.11	0.07	-0.05	-0.30	-0.11		
15	TZ Span >=6 and <9 hours	0.46	0.50	N/A	0.02	0.07	0.11	0.00	0.17	-0.01	-0.06	-0.10	-0.16	0.28	0.35	0.21	-0.42	-0.50	
16	TZ Span >=9 hours	0.32	0.47	N/A	-0.18	0.11	0.05	0.06	-0.27	-0.03	0.07	0.01	0.00	-0.09	0.05	0.61	-0.31	-0.37	-0.62

Absolute value of **r>0.33** is significant at **p<0.001**; *r>0.24* at *p<0.01*; r>0.18 at p<0.05; r>0.15 at p<0.10); All ICC values reported are significant at p<0.001.

project coordinators, business analysts, and systems analysts from multiple departments. We obtained initial surveys from 300 technical project managers, who provided descriptions of their projects and names of 2318 members. 1311 (56.6%) of these members completed the survey and 1039 of them were project managers and project members, which we categorized as *"core"* members. We used data for all projects with 2 or more responses by core members, yielding a total of 123 projects with an average of 10 core members. The remaining 272 contributors were project advisors, outside experts and other stakeholders with limited involvement in the project, who we categorized as *"noncore"* members.

We analyzed the data using hierarchical regression models employing ordinary least squares (OLS) in three regression equations formulated as a path model. In the first equation, we regressed several control and all geographic configuration variables on coordination problems. In the second equation, we regressed the same variables on team performance to evaluate if our variables of interest had a direct effect on team performance. In the third equation, we added coordination problems to the team performance model to test the structural path to team performance with coordination problems as a mediator. To reduce problems of common-method variance, we used data from core members for predictor variables and data from all members (core and noncore) for team performance. We explored several alternative methods to OLS and finally settled with OLS, because none of its assumptions were violated, indicating that the OLS models provide the most efficient and unbiased estimators [46][2] . As prescribed with path models, we included all the variables from the first model in the second model, in order to test every possible path, not just the ones we modeled.

A. Study Variables

The study variables, their descriptive statistics, and the correlation matrix are shown in Table I. All variables were constructed with survey items, except for the two temporal separation variables, which were constructed based on actual time zone differences. The survey items were first evaluated using factor analysis with a varimax rotation and grouped into variables using a criterion of eigenvalues greater than 1. All factors had high factor loadings of 0.67 or higher, and all item groupings had Cronbach-α reliability scores of 0.73 or higher. The respective survey items, reliability scores, and factor loadings are shown

[2]We decided against using LISREL because it is recommended to have at least 150 observations, which is typical of Maximum Likelihood estimation methods, and it requires the data to be rigorously tested for normality [47]; seemingly unrelated regressions (SUR) are more efficient estimators when the residuals of the various models are correlated, but because we have a non-recursive path model with only one intermediate endogenous variable, the residuals between our two OLS equations are not correlated [48] and endogeneity is not a concern [46]; we also decided against two and three stage least squares (2SLS, 3SLS) methods because these methods are only as good as the exogenous instrument variables selected [46] and we could not identify appropriate instruments for coordination problems; finally, we also explored the use of PLS and decided against it because, even though PLS handles relatively smaller sample sizes and does not have strong distributional assumptions, it imposes additional analytical constraints in simultaneously estimating both the measurement and structural models [49].

TABLE II
SURVEY ITEMS

Survey Items	**Factor Loadings**		
Factor Model 1			
Project Resources (reliability α=0.73) (1: not at all, 3: somewhat, 5: very)			
...have available monetary or financial resources	**.753**	.217	-.034
...have available personnel or human resources	**.845**	-.019	.179
...have available information or material resources	**.765**	.029	.276
Project Priority (reliability α=0.85) (1: not at all, 3: somewhat, 5: very)			
...have a very urgent deadline.	-.006	**.889**	.049
...have a high priority status.	.219	**.841**	.120
...require immediate feedback.	.037	**.862**	.107
Task Organization (reliability α=0.74) (1: disagree, 3: neutral, 5: agree)			
...assign roles for who was responsible for what	.186	.178	**.705**
...have substantial agreement on goals, strategies and processes	.307	.053	**.720**
...we distributed responsibilities so that we could work somewhat independently from each other	.003	-.031	**.730**
...we established ground rules, routines and meeting schedules to facilitate our team	.040	.127	**.764**

Factor Model 2	
Communication Tools (α=0.87) (1: rarely, 2: monthly, 3: bi-weekly, 4: weekly, 5: daily)	
...voice communication (e.g. telephone or voice conference)	**.868**
...synchronous text communication (e.g. instant messenger)	**.803**
...asynchronous text communication (e.g. e-mail)	**.874**
...shared desktop based communication (e.g. NetMeeting or WebX)	**.849**

Factor Model 3		
Coordination Problems (reliability α=0.90) (1: disagree, 3: neutral, 5: agree)		
...we often had to re-plan things because of missed delivery dates	**.657**	-.276
...we had many problems due to confusion and misunderstanding (by me or by others) about our task requirements	**.784**	-.140
...we had many problems due to non-adherence (by me or by others) with project processes and procedures	**.764**	-.108
...we had many problems due to priority conflicts	**.728**	-.147
...we had several problems that had to be escalated to higher levels in the organization	**.691**	-.163
...we had many instances of substantial roadblocks	**.741**	-.149
...we had a lot of redundant work	**.722**	-.134
...we had a lot of integration problems	**.769**	-.162
...we had to do a lot of re-work	**.775**	-.184
Team Performance (reliability α=0.81) (1: disagree, 3: neutral, 5: agree)		
...we completed the work on schedule/on-time	-.236	**.828**
...we completed the work well within budget	-.132	**.848**
...the final product met requirements	-.160	**.812**

in Table II. We also tested the predictor variables for discriminant validity in the same model with the predicted variables and found that the predicted variables loaded onto a separate factor, discounting problems with common method bias. The study variables were computed by averaging individual variables for each team. Before aggregation, we computed intra-class correlations (ICC) scores for all the aggregated variables, which are reported in Table I. As recommended to justify aggregation, we evaluated these scores using an F-test to compare within team and between team variances [50]. All the tests were significant at the $p < 0.001$ significance level, indicating more variation between teams than within teams, thus warranting aggregation to the team level. Only core member data was averaged to the team level, except for team performance, which included non-core members to further reduce problems of common method variance in the survey measures. Our main variables of interest in the study include:

1) *Team performance*, measured with survey items inquiring if the project was completed on schedule, on budget and meeting requirements [8], [9].
2) *Coordination problems*, measured with survey items aimed at identifying ineffectively managed technical, temporal, and process dependencies [14].
3) *Spatial separation*, measured for each pair in the team, using a 7-pt scale (1: same room, 2: same hallway, 3: same floor, 4: same building, 5: different building, 6: different city, 7: different country) and then averaged for each team [51]. Our use of this scale is consistent with prior studies that show a stepwise reduction in interaction as a function of geographic location [21].
4) *Number of time zones* represented in each team and the *maximum time zone spanned* by each team, computed based on actual time zone data for each project location.

In addition, we included two coordination process variables:

1) *Coordination via task organization* representing mechanistic coordination [30], [31] involving things like role assignment, division of labor, ground rules and routines. We developed a task organization scale to measure mechanistic coordination using survey items informed by prior research [29].
2) *Coordination via communication,* representing organic coordination, which we measured with survey items that asked about team communication frequency using various methods (e.g., phone, email, conferencing).

We also included in our analysis several control variables. These are variables that prior research has shown to have

TABLE III
REGRESSION ANALYSIS RESULTS

Predictor Variable	Coordination Problems			Team Performance					
				Without Coordination Problems			With Coordination Problems		
	β	p-value	VIF	β	p-value	VIF	β	p-value	VIF
Core Team Size	0.059	0.509	1.88	-0.214	0.052	1.88	-0.175	0.060	1.89
Project Length in Months	-0.078	0.256	1.11	0.047	0.579	1.11	0.006	0.934	1.12
Average Number of Years with Company	-0.011	0.890	1.53	-0.138	0.163	1.53	-0.146	0.080	1.53
Project Resources	-0.217	0.007	1.48	0.298	0.003	1.48	0.152	0.073	1.58
Project Priority	0.356	<0.001	1.27	-0.035	0.695	1.27	0.203	0.016	1.53
Coordination via Task Organization	-0.535	<0.001	1.38	0.233	0.014	1.38	-0.126	0.184	1.98
Coordination via Communication Tools	-0.024	0.781	1.70	-0.070	0.504	1.70	-0.085	0.329	1.70
Spatial separation	-0.021	0.804	1.60	0.040	0.695	1.60	0.026	0.760	1.60
Number of Time Zones	-0.053	0.578	2.10	0.107	0.356	2.10	0.072	0.462	2.10
Maximum Time Zone Span	0.164	0.034	1.37	-0.230	0.015	1.37	-0.121	0.133	1.42
Coordination Problems							-0.669	0.000	2.10
R^2 (explained variance)	0.523			0.291			0.505		
P-Value (predictive power)	<0.001			<0.001			<0.001		
P-Value increase with Coordination Problems							<0.001		
Condition Index (CI)	2.91			2.91			3.02		

Note: the β values in the table are standardized regression coefficients.

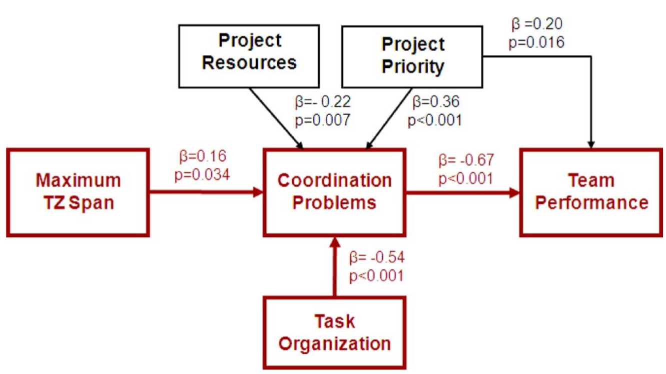

Fig. 3. Results.

an effect on project coordination and performance. They include core team size, project length, team members' average number of years with the company, project resources, and project priority.

IV. DATA ANALYSIS AND RESULTS

The main regression analysis results and colinearity statistics are shown in Table III. The significant effects are illustrated in Fig. 3. We used this regression model to test our hypotheses. Because the maximum time zone variable was significant in the main model, we also formulated a "fixed effects" model [52] by grouping our observations into four categories of maximum time zone span to further investigate, which span magnitude (3–6, 6–9 or 9–12 hours), had the strongest impact as compared to the baseline (0–3 hours, i.e., little or no time difference). We used three binary variables with a value of 1 if the time zone span was 3–6, 6–9, and 9–12 hours, respectively, and 0 otherwise. The (fourth) omitted binary variable for the 0–3 hours span category is the baseline group represented by the regression intercept. The corresponding results for the fixed effects model are shown in Table V.

While the correlation between our spatial separation measured with time separation variables was only moderate (0.40 with the number of time zones and 0.20 with maximum time zone difference spanned by the team), colinearity is always a concern with these variables so we tested the model using the condition index (CI) [53] and variance inflation factors (VIF) [54]. The CI measures the overall linear relationships with a value over 30 indicating high colinearity for the model as a whole. The VIFs measure how much each independent variable contributes to colinearity. High VIFs of 10 or more indicate that an independent variable may be linearly dependent on others, pointing to colinearity problems. Because the CI can sometimes be inflated by colinear relations with the regression constant, we centered all variables with respect to their means so that the intercept of the regression model goes through the origin, thus eliminating colinearity with the constant. Because our models do not include second order terms, the regression coefficients, standard errors, and VIFs are identical to the uncentered regression models. As Table III shows, our models—CIs and variable VIFs —were substantially below these recommended thresholds indicating no problems with multi-colinearity.

In addition to the variables discussed above, we tested a number of control variables that could affect coordination and performance (e.g., project resources, project priority, task uncertainty, task type, team experience), and removed nonsignificant control variables based on an F-test to preserve degrees of freedom. Project resource was negatively associated with coordination problems ($\beta = -0.22$, $p = 0.007$), whereas project priority was positively associated with coordination problems ($\beta = 0.36$, $p < 0.001$). Interestingly, when the coordination problems variable was introduced as a mediator, the direct effect of project priority

TABLE IV
DIRECT, INDIRECT, AND TOTAL EFFECTS ON COORDINATION AND TEAM PERFORMANCE

Predictor Variable	Coordination Problems			Team Performance				
	Direct Effect	Spurious	ρ	Direct Effect	Indirect Effect	Total	Spurious	ρ
Core Team Size	0.07	0.05	0.12	-0.18	-0.05	-0.23	0.07	-0.16
Project Length in Months	-0.08	0.00	-0.08	0.01	0.05	0.06	-0.04	0.02
Average Number of Years with Company	-0.01	-0.04	-0.05	-0.15	0.01	-0.14	0.09	-0.05
Project Resources	-0.22	-0.17	-0.39	0.15	0.15	0.30	0.12	0.42
Project Priority	0.36	-0.14	0.22	0.20	-0.24	-0.04	0.09	0.05
Coordination via Task Organization	-0.54	-0.05	-0.59	-0.13	0.36	0.23	0.16	0.39
Coordination via Communication Tools	-0.02	0.03	0.01	-0.09	0.01	-0.08	0.11	0.03
Spatial separation	-0.02	-0.06	-0.08	0.03	0.01	0.04	-0.04	0.00
Number of Time Zones	-0.05	0.13	0.08	0.07	0.03	0.10	-0.18	-0.08
Maximum Time Zone Span	0.16	0.04	0.20	-0.12	-0.11	-0.23	0.03	-0.20
Coordination Problems				-0.67		-0.67	0.03	

TABLE V
REGRESSION ANALYSIS RESULTS WITH TIME SEPARATION CATEGORIES

Predictor Variable	Coordination Problems			Team Performance					
				Without Coordination Problems			With Coordination Problems		
	β	p-value	VIF	β	p-value	VIF	β	p-value	VIF
Core Team Size	0.060	0.508	1.90	-.219	.048	1.89	-.178	.053	1.89
Project Length in Months	-0.067	0.341	1.15	.059	.487	1.14	.014	.849	1.14
Average Number of Years with Company	-0.032	0.699	1.59	-.159	.117	1.59	-.181	.033	1.59
Project Resources	-0.233	0.005	1.54	.277	.006	1.54	.118	.170	1.66
Project Priority	0.374	<0.001	1.34	-.019	.841	1.34	.238	.006	1.63
Coordination via Task Organization	-0.534	<0.001	1.39	.244	.011	1.39	-.121	.197	2.00
Coordination via Communication Tools	-0.019	0.824	1.72	-.063	.550	1.71	-.076	.384	1.71
Spatial separation	-0.022	0.791	1.61	.019	.851	1.61	.004	.963	1.61
Number of Time Zones	-0.069	0.480	2.22	.079	.505	2.22	.032	.746	2.23
Maximum Time Zone Span 3 hrs or more to 6 hrs	0.096	0.233	1.50	.031	.753	1.50	.097	.238	1.51
Maximum Time Zone Span 6 hrs or more to 9 hrs	0.232	0.042	2.96	-.121	.379	2.96	.038	.744	3.07
Maximum Time Zone Span 9 hrs or more	0.239	0.024	2.53	-.286	.026	2.53	-.122	.259	2.65
Coordination Problems							-.685	.000	2.11
R^2 (explained variance)	0.527			0.300			0.522		
P-Value (predictive power)	<0.001			<0.001			<0.001		
P-Value increase with Coordination Problems							<0.001		
Condition Index (CI)	6.116			6.116			6.323		

Note : the β values in the table are standardized regression coefficients.

on team performance became positive and significant ($\beta = 0.20$, $p = 0.016$).

In support of Hypothesis 1a, we found that task organization had a very strong negative effect on coordination problems ($\beta = -0.54$, $p < 0.001$). Surprisingly, we did not find evidence to support Hypothesis 1b that communication tools have an effect on coordination problems. Prior evidence has found that team communication helps coordination within team dyads [38], but our results suggest that for the team as a whole, task organization is more effective in helping reduce coordination problems.

Controlling for time separation variables, we did not find that spatial separation had an effect on coordination problems, thus failing to support Hypothesis 2a. Many studies have found that spatial separation affects coordination, but some of the same studies have also found that the effects of spatial separation are mediated by other factors like team size (i.e., multiple site teams tend to be larger) [17] or argue that the effects of spatial separation are confounded with other factors like time zones [18]. Also, given the fact that the organization we studied is very global with many years of experience working across geographic locations, we were not entirely surprised to find that coordination was not affected by spatial separation.

We also did not find support for Hypothesis 2b, that the number of time zones on a team affects coordination problems.

However, in support of Hypothesis 2c, we found that the maximum time zone spanned by a team increased coordination problems ($\beta = 0.16$, $p = 0.034$). This suggests that coordination problems may have occurred mainly because of the reduced overlap in work hours. While we found no significant difference between the 0–3 and 3–6 hours spans in the fixed effects model, we did find that, compared to the 0–3 hspan group, the 6–9 hours span ($\beta = 0.232$, $p = 0.042$) and 9–12 hours span ($\beta = 0.239$, $p = 0.024$) groups had more coordination problems. These results confirm that small time separation with some overlapping hours do not create substantial coordination problems, but greater time zone span is associated with more coordination problems, with the 9–12 hours span having the most severe effect.

This is exemplified by the following comment made by another global team member we interviewed: *"It's difficult when you're in a high stress portion of the project. . .the release date is coming and all that kind of stuff, so last night we had meeting, 5 pm my time, 8 pm their time, and I said, look, at the end of your day, make sure you send me an e-mail because we won't be able talk until Tuesday. Make sure you send me an e-mail what you've got done and what you need me to do between now and my Sunday, your Monday. They sent me an e-mail, but it had a ton of things like subtle questions that I would have asked, subtle things that I don't understand, and I can't call them at home on their Saturday morning, at 3:00 in the morning I can't go call them."*

A. Explaining Team Performance

As we show in Table III, we first regressed on team performance the same independent variables we used in the coordination models. We then added the coordination problems variable into the model. This method allows us to investigate mediation effects by observing if the independent variables have direct effects on team performance when the coordination problems variable is omitted, and whether these effects go away or become less significant when the coordination problems variable is added [55]. In the first model without coordination problems, we found that maximum time zone span had a negative and statistically significant association with team performance ($\beta = -0.23$, $p = 0.015$). Interestingly, spatial separation alone was not directly associated with team performance, even after removing the maximum time zone span variable from the model. We also found in the fixed effects model that only the 9–12 hours span group had significantly lower performance compared with the 0–3 hours time span group ($\beta = -0.286$, $p = 0.026$), confirming once again that teams with larger time zone spans had lower performance, but that smaller time zone spans were not as problematic. However, as we discuss next, this negative association between time zone span and team performance disappears when we account for coordination problems experienced by the team.

Consistent with our predictions, when we included coordination problems in the analysis, the direct association between maximum time zone span and team performance disappeared, but the negative effect of coordination problems on team performance was significant ($\beta = -0.67$, $p < 0.001$) suggesting that coordination problems mediate the effect of maximum time zone span on team performance. To test this mediation effect more formally, we modeled team performance with and without the coordination problems variable as a predictor [55] and then conducted a Sobel mediation test, which yielded a statistically significant effect ($t = -2.096$, $p = 0.036$) confirming that the negative effect of time zone span on team performance is indeed mediated by coordination. Table IV also shows the direct, indirect, and spurious effects [27] of the geographic configuration and other antecedent, and coordination problems on team performance. As the table shows, the spurious effect of maximum time zone span on team performance is only 0.03, compared to a direct and indirect β path coefficients of -0.12 and -0.11, respectively, further supporting our mediation results.

Overall, our results provide sound empirical evidence that lower performance was due primarily to coordination problems created by a large time zone span. The important implication of this finding is that when team coordination is taken care of with effective task organization (i.e., role assignments, division of labor and responsibilities, work rules and routines) the negative effects of maximum time zone span on team performance can be mitigated to some extent. Naturally, since we did not find direct effects of spatial separation or number of time zones on coordination problems or team performance, we could not test the mediation effects of coordination problems for these variables. Therefore, we found support for Hypothesis 3c, but not for 3a and 3b. As we discuss below, these results suggest that teams have learned to bridge spatial and time separation through technology, except when the time zone span is very large.

V. Discussion

Our study has important implications for research and practice. For research, our study is one of the first to jointly study the effect of spatial separation and more than one parameter of time separation. Our study is also the first to find that the effect of time separation is mediated by coordination problems. For practice, our study was partly motivated by the need to use academic research to influence the design of collaboration tools to support geographically dispersed teams. We discuss these contributions in more detail next.

A. Implications for Research

Overall, the most important aspect of our findings is the path we found from time separation to team performance, which had not been demonstrated in prior research. Consistent with our expectations, the maximum time zone spanned by the team had a substantial positive effect on coordination problems. Also consistent with our predictions, time zone span had a detrimental effect on team performance, and this effect was indirect and mediated by coordination problems.

We also found that task organization helped reduce coordination problems, thus helping offset coordination problems that may arise due to large time zone spans represented in the team. Once again, our study represents a contribution to the research literature, because it is the first study to find the impact of task organization on team coordination and performance. This is important for research because our results suggest that team level coordination problems go beyond dyadic issues. Any two members can coordinate their work across time zones by communicating, but general task organization is necessary for the full team to operate effectively in geographically dispersed contexts. Given the importance of task organization for reducing coordination problems, we investigated if geographic configuration influenced the selection of particular coordination mechanisms. We were surprised that none of the geographic configuration variables we examined influenced the use of task organization. Global team members appear to select a mix of coordination processes based on personal preferences rather than geographic configuration factors. This provides opportunities for further research on the antecedents of coordination processes.

We were somewhat surprised that team communication did not help reduce coordination problems. The interesting thing to note is that, as we mentioned earlier, prior studies found team communication to help reduce coordination delays in dyads [38], but our study shows that it does not help alleviate coordination problems at the team level. Conversely, task organization (i.e., mechanistic coordination) does not operate at the dyad level, but it has the strongest effect on team level coordination outcomes. These results contribute to the research literature because they provide nuanced explanations for when communication or task organization may be effective for coordination, thus informing future research. Our results suggest that team members should communicate selectively in pairs as needed to resolve any coordination delay issues with their peers, but for the team's coordination as a unit, task organization is fundamental.

We were also surprised that our predictions for the impact of spatial separation and number of time zones were not supported. We speculated that perhaps we might have needed to control for other variables that could affect coordination, such as project complexity. Because the projects we studied involved inherently complex technical work, it was difficult to develop a single complexity metric to analyze all projects. For example, it is difficult to compare the complexity of a flash memory design project with the complexity of a business application development project. Given the absence of a standard metric of complexity applicable to all projects, we used several other survey variables associated with task complexity (e.g., task uncertainty, type of task, team composition). However, we found no systematic differences across spatial separation or time separation variables, and when we modeled these variables in our regression models they were not significant. We conjecture that organizations with substantial experience working globally, like the company in which we conducted this study, have learned to work across distance and have learned to bridge spatial separation effectively with technology. Similarly, working across multiple time zones does not appear to be a problem for these organizations, provided that the magnitude of time zone spanned by the team is not very large. Future research needs to control for the various geographic configuration variables that may be at play and also control for the organization's experience working in the particular geographic context under study.

B. Implications for Practice

In a nutshell, reducing coordination problems is key to achieving high performance in teams separated by many time zones. Naturally, coordination problems can be minimized by reducing the time zone difference spanned by the team, but when reducing the time zone span is not feasible or practical, organizing project tasks more effectively can mitigate coordination problems. Interestingly, when we presented these results to managers at the company we studied, they were consistently surprised that task organization had a greater impact on coordination problems than communication tools when teams had time separation. This suggests that common intuition holds that technology is the primary answer for challenges faced in global teams when, in fact, there are still important fundamentals to consider such as organizing and dividing up work most effectively. This has important implications for global project managers. Implementing policies that foster the adoption of effective task organization practices can pay off by reducing coordination problems. Also, since the mix of communication options available to the team is more severely reduced when team members span larger time zone differences, task organization can be most effective because teams can coordinate with less direct communication.

Because spatial separation and time separation were hypothesized to hinder coordination, we anticipated that both would foster greater use of communication tools and task organization to coordinate work. But we found that neither spatial separation nor time separation predicted greater use of either, suggesting that coordination methods are more a matter of choice rather than based on rational decisions about what may work better for particular geographic configuration contexts. Similarly, we speculated that the beneficial effects of task organization on coordination would be moderated by time separation (i.e., that task organization would have a stronger effect for teams with more time zone span), but we did not find a significant interaction effect. However, the fact that large time zone spans create coordination problems, and that task organization offsets these problems provides a strong argument for the importance of task organization as time zone span increases.

Finally, our results also suggest that coordination problems can be reduced by making more (financial, personnel, and information) resources available to the project, which would help improve team performance directly. It is important to note that our variable measures availability and not use of resources, suggesting that projects that are well supported financially or otherwise exhibit better coordination and performance. In contrast, project priority seems to create coordination problems, perhaps because an increased sense of urgency may cause the team to operate haphazardly, but this sense of high priority appears to help meet schedules, budgets, and system requirements.

C. Implications for Collaboration Tool Design

Our study also provides insights for collaboration tool designers to help teams organize their workflow and manage tasks effectively. Tools can be configured so that the most appropriate task organization processes are evident to team members. For example, the use of workflow automation to track status and make workflow more explicit may reduce the amount of effort needed to coordinate task activities and reduce the number of real-time meetings needed to coordinate work. Collaboration tools can also be designed to help teams figure out effective work schedule shifting schemes in order to help them reduce the effective maximum time zone span. Shifting schedules to reduce time zone span is very easy when a few time zones are involved, but it can be a daunting task with many time zones, which is where collaboration tools can help.

Collaboration tools can also be designed to help team members and project managers select an optimal distribution of task dependencies across sites based on the project resources available at those sites, thus reducing the coordination overhead imposed on teams. When task dependencies across sites are minimized, members at one site can work more independently. But tools could also provide explicit representation and tracking of such task dependencies. A visual representation of who is doing what and how various tasks relate to each other, can keep the team more organized; it can also foster knowledge sharing because members can figure out more easily who to contact when dealing with task dependency issues. Finally, given that task organization is the strongest predictor of reduced coordination problems, collaboration tools designed to provide organization and structure in a project are generally beneficial.

VI. Limitations and Conclusion

Our study is not without limitations. First, we conducted the study with a single organization, so our results may have been influenced by its idiosyncrasies. Nevertheless, this organization is a large semiconductor manufacturing firm with substantial experience working with technical teams across geographically dispersed locations. Therefore, the validity of our results can be generalized to similar technology companies working globally. Also, because the study was conducted with survey methods, our results are subject to the usual limitations typical of survey studies. Nevertheless, we took good precautions to ensure that our survey methods were rigorous, including grouping variables using factor loadings, provisions for common method variance, and statistical justification for aggregation of individual variables to the team level, among other things.

Despite these limitations, our study represents an important contribution to the research literature. Understanding how time separation, not simply spatial separation, affects team coordination and performance is critical to success in global technical projects. In cases where geographic configuration can be altered, project managers should strive to reduce time zone spans. When altering the geographic configuration of a team is not an option, project managers should strive to decouple dependencies across time zones to make the work more independent, and manage the remaining dependencies with effective task organization mechanisms and configurable collaboration tools aimed at reducing coordination problems.

Our study provides a unique contribution to the literature in a number of ways. It is the first study to tease out the effect of time separation from spatial separation at the team level. It also uses different geographic configuration measures (i.e., time separation, number of time zones, and maximum time zone span) to find the differential effects of geographic configuration on coordination problems and team performance, therefore, underscoring the importance of decomposing geographic configuration variables into finely grained components. The study finds that the effect of a geographic configuration variable—time zone span—on performance is indirect and mediated by coordination problems, thus highlighting the importance of coordination in geographically dispersed technical tasks. Finally, the research provides evidence supporting the importance of mechanistic coordination in the form of task organization in helping offset some of the detrimental effects of time zone span on coordination problems.

References

[1] M. Gothe, K. Fryer, "Global software development and delivery: Trends and challenges," IBM Developer Works, January 15, 2008.

[2] F. Karamouzis, "Global delivery model market competitiveness cube provides insights into IT services providers' direction," Stamford, CT: Gartner Research Report: G00171213, Sep. 29, 2009.

[3] J. G. Cooprider and J. C. Henderson, "Technology-process Fit: Perspectives on achieving prototyping effectiveness," *J. Manage. Inf. Syst.*, vol. 7, no. 3, pp. 67–87, Winter 1990/1991.

[4] S. R. Nidumolu, "The effect of coordination and uncertainty on software project performance: Residual performance risk as an intervening variable," *Inf. Syst. Res.*, vol. 6, no. 3, pp. 191–219, Sep. 1995.

[5] B. H. Wixom and H. J. Watson, "An empirical investigation of the factors affecting data warehousing success," *MIS Q.*, vol. 25, pp. 17–41, Mar. 2001.

[6] C. Deephouse, T. Mukhopadhyay, D. R. Goldenson, and M. I. Keller, "Software processes and project performance," *J. Manage. Inf. Syst.*, vol. 12, pp. 187–205, Dec. 1995.

[7] R. F. Powers and G. W. Dickson, "MIS project management: Myths, opinions, and reality," *Calif. Manage. Rev.*, vol. 15, no. 3, pp. 147–156, 1973.

[8] W. H. DeLone and E. R. McLean, "The DeLone and McLean model of information systems success: A ten-year update," *J. Manage. Inf. Syst.*, vol. 19, pp. 9–30, 2003.

[9] D. Ancona and D. Caldwell, "Demography and design: Predictors of new product team performance," *Org. Sci.*, vol. 3, pp. 321–341, 1992.

[10] T. Malone and K. Crowston, "The interdisciplinary study of coordination," *ACM Comput. Surv.*, vol. 26, pp. 87–119, 1994.

[11] C. Bullen and J. Bennett, "Groupware in practice: An interpretation of work experiences," in *Groupware and Computer-Supported Cooperative Work: Assisting Human-Human Collaboration*, R. Baecker, Ed. San Francisco, CA: Morgan Kaufman, 1993, pp. 69–84.

[12] J. McGrath and A. Hollingshead, *Groups Interacting With Technology*. Newbury Park, CA: Sage, 1994.

[13] C. D. Cramton, "The mutual knowledge problem and its consequences for dispersed collaboration," *Org. Sci.*, vol. 12, pp. 346–371, 2001.

[14] J. A. Espinosa, S. A. Slaughter, R. E. Kraut, and J. D. Herbsleb, "Team knowledge and coordination in geographically distributed software development," *J. Manage. Inf. Syst.*, vol. 24, pp. 135–169, 2007.

[15] R. E. Grinter, J. D. Herbsleb, and D. E. Perry, "The geography of coordination: Dealing with distance in R&D work," presented at the 1999 Int. ACM SIGGROUP Conf. Supporting Group Work (Group 99), Phoenix, AZ.

[16] J. D. Herbsleb and R. E. Grinter, "Architectures, coordination, and distance: Conway's law and beyond," *IEEE Softw.*, vol. 16, no. 5, pp. 63–70, Sep./Oct. 1999.

[17] J. D. Herbsleb and A. Mockus, "An empirical study of speed and communication in globally distributed software development," *IEEE Trans. Softw. Eng.*, vol. 29, no. 6, pp. 481–494, Jun. 2003.

[18] G. M. Olson and J. S. Olson, "Distance Matters," *Human-Comput. Interact.*, vol. 15, pp. 139–179, 2000.

[19] D. J. Armstrong and P. Cole, "Managing distances and differences in geographically distributed work groups," in *Distributed Work*, P. Hinds and S. Kiesler, Eds. Cambridge, MA: MIT Press, 2002, pp. 187–215.

[20] E. F. McDonough, K. Kahn, and G. Barczak, "An investigation of the use of global, virtual, and collocated new product development teams," *J. Prod. Innovat. Manage.*, vol. 18, pp. 110–120, 2001.

[21] T. J. Allen, *Managing the Flow of Technology*. Cambridge, MA: MIT Press, 1977.
[22] R. E. Kraut, C. Egido, and J. Gallagher, "Patterns of contact and communication in scientific research collaboration," in *Intellectual Teamwork: Social and Technological Foundations of Cooperative Work*, R. E. Kraut, C. Egido, and J. Gallagher, Eds. Hillsdale, NJ: Lawrence Erlbaum Associates, 1990, pp. 149–171.
[23] K. Sobel-Lojeski, *Leading the Virtual Workforce: How Great Leaders Transform Organizations in the 21st Century*. New York: Wiley, 2009.
[24] M. E. Sosa, S. D. Eppinger, M. Pich, D. G. McKendrick, and S. K. Stout, "Factors that influence technical communication in distributed product development: An empirical study in the telecommunications industry," *IEEE Trans. Eng. Manage.*, vol. 49, no. 1, pp. 45–58, Feb. 2002.
[25] M. B. O'Leary and J. N. Cummings, "The spatial, temporal, and configurational characteristics of geographic dispersion in teams," *MIS Q.*, vol. 31, pp. 433–452, 2007.
[26] E. Carmel, "Building your information systems from the other side of the world: How Infosys manages time zone differences," *MIS Q. Execut.*, vol. 5, pp. 43–53, 2006.
[27] M. B. Pinto, J. K. Pinto, and J. E. Prescott, "Antecedents and consequences of project team cross-functional cooperation," *Manage. Sci.*, vol. 39, pp. 1281–1297, 1993.
[28] S. Faraj and L. Sproull, "Coordinating expertise in software development teams," *Manage. Sci.*, vol. 46, pp. 1554–1568, 2000.
[29] A. H. Van de Ven, L. A. Delbecq, and R. J. Koenig, "Determinants of coordination modes within organizations," *Amer. Sociol. Rev.*, vol. 41, pp. 322–338, 1976.
[30] J. March and H. A. Simon, *Organizations*. New York: Wiley, 1958.
[31] J. Thompson, *Organizations in Action*. New York: McGraw-Hill, 1967.
[32] R. E. Grinter, "Workflow systems: Occasions for success and failure," *Comput. Supported Cooperat. Work*, vol. 9, pp. 189–214, 2000.
[33] R. E. Kraut and L. A. Streeter, "Coordination in software development," *Commun. ACM*, vol. 38, pp. 69–81, 1995.
[34] J. A. Espinosa, J. N. Cummings, J. M. Wilson, and B. M. Pearce, "Team boundary issues across multiple global firms," *J. Manage. Inf. Syst.*, vol. 19, pp. 157–190, 2003.
[35] W. Orlikowski, "Knowing in practice: Enacting a collective capability in distributed organizing," *Org. Sci.*, vol. 13, pp. 249–273, 2002.
[36] M. B. Watson-Manheim, K. Chudoba, and K. Crowston, "Discontinuities and continuities: A new way to understand virtual work," *Inf., Technol. People*, vol. 15, pp. 191–209, 2002.
[37] S. Kiesler and J. N. Cummings, "What do we know about proximity in work groups? A legacy of research on physical distance," in *Distributed Work*, P. Hinds and S. Kiesler, Eds. Cambridge, MA: MIT Press, 2002, pp. 57–80.
[38] J. Cummings, J. A. Espinosa, and C. Pickering, "Crossing spatial and temporal boundaries in globally distributed projects: A relational model of coordination delay," *Inf. Syst. Res.*, vol. 20, pp. 420–439, 2009.
[39] E. Carmel and P. Tjia, *Offshoring Information Technology: Sourcing and Outsourcing to a Global Workforce*. Cambridge, U.K.: Cambridge Univ. Press, 2005.
[40] J. A. Espinosa and C. Pickering, "The effect of time separation on coordination processes and outcomes: A case study,". (2006). presented at 39th Hawaiian Int. Conf. Syst. Sci., Poipu, Kauai, HI, 2006.
[41] R. E. Wood, "Task complexity: Definition of the construct," *Org. Behav. Human Decis. Process.*, vol. 37, pp. 60–82, 1986.
[42] J. A. Espinosa, S. A. Slaughter, R. E. Kraut, and J. D. Herbsleb, "Familiarity, complexity and team performance in geographically distributed software development," *Org. Sci.*, vol. 18, pp. 613–630, 2007.
[43] J. D. Herbsleb, A. Mockus, T. Finholt, and R. E. Grinter, "Distance, dependencies and delay in a global collaboration," presented at 2000 ACM Conf. Comput. Support. Cooperat. Work, Philadelphia, PA.
[44] J. Herbsleb, A. Mockus, T. Finholt, and R. E. Grinter, "An empirical study of global software development: Distance and speed,". (2001). presented at 23rd. Int. Conf. Softw. Eng. (ICSE), Toronto, Canada, 2001.
[45] E. Carmel, *Global Software Teams*. Upper Saddle River, NJ: Prentice-Hall, 1999.
[46] W. Greene, *Econometric Analysis*. Upper Saddle River, NJ: Prentice-Hall, 1997.
[47] D. Gefen, D. W. Straub, and M.-C. Boudreau, "Structural equation modeling and regression: Guidelines for research and practice," *Commun. AIS*, vol. 4, pp. 1–77, 2000.
[48] P. Tait and I. Vessey, "Authors' response to bordoloi and lauer: Problems of applying OLS/path analysis for estimating structural (multi-equation) models," *MIS Q.*, vol. 13, p. 387, 1989.
[49] G. A. Marcoulides and C. Saunders, "PLS: A silver bullet?," *MIS Q.*, vol. 30, pp. iv–viii, 2006.
[50] D. A. Kenny and L. LaVoie, "Interpersonal relations and group processes," *J. Personal. Soc. Psychol.*, vol. 48, pp. 339–348, 1985.
[51] J. Cummings, "Work groups, structural diversity, and knowledge sharing in a global organization," *Manage. Sci.*, vol. 50, pp. 352–364, 2004.
[52] P. Kennedy, *A Guide to Econometrics*. Cambridge, MA: MIT Press, 1992.
[53] D. Belsley, E. Kuh, and R. Welsch, *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. New York: Wiley, 1980.
[54] D. W. Marquardt, "Generalized inverses, ridge regression, biased linear estimation, and non-linear estimation," *Technometrics*, vol. 12, pp. 591–612, 1970.
[55] R. Baron and D. Kenny, "The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations," *J. Personal. Soc. Psychol.*, vol. 51, pp. 1173–1182, 1986.

J. Alberto Espinosa (M'11) received the Ph.D. degree in information systems from the Tepper School of Business at Carnegie Mellon University, Pittsburgh, PA, in 2002.

He is an Associate Professor of Information Technology at the Kogod School of Business, American University, Northwest Washington, DC. His research focuses on coordination and performance in global technical projects across global boundaries, particularly spatial and temporal distance. His current research areas include coordination of technical work across time zones and coordination in large-scale technical collaboration tasks like enterprise architecture.

Jonathon N. Cummings received the Ph.D. degree from Carnegie Mellon University, Pittsburgh, PA, in 2001.

He is an Associate Professor of Management at the Fuqua School of Business, Duke University, Durham, NC. His research focuses on social networks and teams in corporations and science, including the role of knowledge sharing in work distributed across different geographic locations. He is an Associate Editor of MANAGEMENT SCIENCE and Faculty Director of the Center for Technology, Entertainment, and Media (CTEM) at Fuqua.

Cindy Pickering received the B.S.E.E. degree from Pennsylvania State University, University Park, PA, in 1981.

She is a Senior Principal Engineer for Intel IT Strategy, Architecture and Innovation, and Director of the Chandler, Arizona Collaboration Research Lab, AZ. She has 30 years of industry research experience. Her research focuses on global team collaboration combining needs assessment, new concept development, emerging usage models definition, and enabling architecture and technology exploration.